

2 Data presentation and interpretation	2.1	Interpret diagrams for single-variable data, including understanding that area in a histogram represents frequency. Connect to probability distributions.	Students should be familiar with histograms, frequency polygons, box and whisker plots (including outliers) and cumulative frequency diagrams.
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Topics	What students need to learn:		
		Content	Guidance
2 Data presentation and interpretation <i>continued</i>	2.2	Interpret scatter diagrams and regression lines for bivariate data, including recognition of scatter diagrams which include distinct sections of the population (calculations involving regression lines are excluded). Understand informal interpretation of correlation. Understand that correlation does not imply causation.	Students should be familiar with the terms explanatory (independent) and response (dependent) variables. Use of interpolation and the dangers of extrapolation. Variables other than x and y may be used. Use to make predictions within the range of values of the explanatory variable. Change of variable may be required, e.g. using knowledge of logarithms to reduce a relationship of the form $y = ax^n$ or $y = kb^x$ into linear form to estimate a and n or k and b. Use of terms such as positive, negative, zero, strong and weak are expected.
	2.3	Interpret measures of central tendency and variation, extending to standard deviation. Be able to calculate standard deviation, including from summary statistics.	Data may be discrete, continuous, grouped or ungrouped. Understanding and use of coding. Measures of central tendency: mean, median, mode. Measures of variation: variance, standard deviation, range and interpercentile ranges. Use of linear interpolation to calculate percentiles from grouped data is expected. Students should be able to use the statistic x $S_{xx} = \sum (x - \bar{x})^2 = \sum x^2 - \frac{(\sum x)^2}{n}$ Use of standard deviation = $\sqrt{\frac{S_{xx}}{n}}$ (or equivalent) is expected but the use of $\sqrt{\frac{S_{xx}}{n-1}}$ is not.

Topics	What students need to learn:		
	Content		Guidance
2 Data presentation and interpretation <i>continued</i>	2.4	Recognise and interpret possible outliers in data sets and statistical diagrams. Select or critique data presentation techniques in the context of a statistical problem. Be able to clean data, including dealing with missing data, errors and outliers.	Any rule needed to identify outliers will be specified in the question. For example, use of $Q_1 - 1.5 \times \text{IQR}$ and $Q_3 + 1.5 \times \text{IQR}$ or $\text{mean} \pm 3 \times \text{standard deviation}$. Students will be expected to draw simple inferences and give interpretations to measures of central tendency and variation. Significance tests, other than those mentioned in Section 5, will not be expected. For example, students may be asked to identify possible outliers on a box plot or scatter diagram.